Hyeonbeen (Edward) Lee

edward.hyeonbeen.lee@gmail.com • GitHub (@leehyeonbeen) • Google Scholar

EDUCATION

Kyung Hee University Mar. 2022 — Feb. 2024
M.S. in Mechanical Engineering (Advisor: Jin-gyun Kim) GPA: 4.10/4.3
Thesis: *'Composite neural network with differential propagation for modeling impulsive nonlinear dynamic systems'*
Coursework: Probability & Statistics, Advanced linear algebra, ROS, Computer vision & Artificial intelligence, Data analysis

Kyung Hee University (*Absence for military service in May 2017 — Feb. 2019*) Mar. 2015 — Feb. 2022
B.S. in Mechanical Engineering (Advisor: Jin-gyun Kim, Shinkyu Jeong) (Upper Div.: 4.04/4.3) GPA: 3.60/4.3
Thesis: *'Data-driven aerodynamic coefficient prediction using deep neural network and PARSEC airfoil parameterization'*
Coursework: Computational dynamics, Robotics, Convex optimization & Deep learning, Numerical analysis, Finite element methods, Algorithm & programming, Materials science, Thermodynamics, Fluid mechanics

RESEARCH INTEREST

Robot learning, Sequential modeling, Sim2Real, Multimodal perception, Physics-informed learning

PUBLICATIONS

3. **H. Lee**, M. Jung, J.G. Kim. (2025), "Forecasting high-frequency robot joint forces in contact-rich manipulation via short-time stationarity and neural frequency filters", *Manuscript in preparation.*
2. **H. Lee**, S. Han, H.S. Choi, J.G. Kim (2024), "cNN-DP: Composite neural network with differential propagation for impulsive nonlinear dynamics", *Journal of Computational Physics (WoS IF Top 2.5% in Physics, Mathematical)*, 112578. [link] [GitHub]
1. S. Han, G.E. Jeong, **H. Lee**, W.S. Choi, J.G. Kim (2023), "Multi-body dynamics model for spent nuclear fuel transportation system under normal transport test conditions", *Nuclear Engineering and Technology (WoS IF Top 13.7% in Nuclear Science & Technology)*, 55(11), 4125-4133. [link]

RESEARCH EXPERIENCES

Kyung Hee University Postgraduate Researcher
Modeling & Simulation Lab., advised by *Jin-gyun Kim* Mar. 2024 — Present
- Developed a scalable framework for forecasting high-frequency forces of robots in contact-rich manipulation scenarios.
- **Ongoing:** Finalizing a manuscript for publication.

KAIST Graduate School of AI Research Engagement
Cognitive Learning for Vision & Robotics Lab., advised by *Joseph J. Lim* Sep. 2023 — Nov. 2023
- Conducted a two-month intensive research review on skill prior regularized reinforcement learning (SPiRL), and delivered a 90-minute technical presentation to lab members and faculty.
- Received a formal internship offer from the lab director based on the presentation performance (offer declined).

Kyung Hee University M.S. Student
Modeling & Simulation Lab., advised by *Jin-gyun Kim* Mar. 2022 — Feb. 2024
- Proposed a novel neural network architecture to model nonlinear dynamic systems with strong local oscillations. Resulted in a first-author publication in *Journal of Computational Physics.*
- Accurately modeled real-world vehicle dynamics via neural metamodeling and high-fidelity simulation for a national research grant by KAERI. Resulted in a co-authored publication in *Nuclear Engineering and Technology.*
- Designed a generative model to estimate real-time, contact-rich joint forces of a marine excavation robot for a national grant by KRISO.
- Actively explored and implemented methods from diverse ML domains, including robot learning, time-series forecasting, computer vision, physics-informed learning and system identification.

Kyung Hee University Undergraduate Researcher
Modeling & Simulation Lab., advised by *Jin-gyun Kim* Jan. 2021 — Feb. 2022
- Investigated data-driven methods for flexible multi-body dynamics as part of a national research grant on vehicle dynamics simulation.

AWARDS & SCHOLARSHIPS

- **Excellent Paper Award** — Korean Society of Mechanical Engineers, Aug. 2023
- **RA Scholarship** — Graduate School of Engineering, Kyung Hee University, Sep. 2022 — Feb. 2023
- **Academic Excellence Scholarship** — Dept. of Mechanical Eng., Kyung Hee University, Mar. 2021

CONFERENCES

J. Han, J.B. Han, S.S. Kim, M.H. Kim, Y.H. Kim, **H. Lee**, J.G. Kim, T.K. Yeu. "Digital twin model of underwater construction robot for real-time grinding simulation", *7th International Conference on Multibody System Dynamics.* [link] — Jun. 9 2024, Madison, WI, USA

H. Lee, J. Han, T. Yeo, J.G. Kim. "Real-time multi-horizon reaction force forecasting of marine robot using interpretable Transformer", *Annual Conference, Korean Society of Mechanical Engineers* (Oral Presentation). [link] — Nov. 1 2023, Incheon, South Korea

H. Lee, S. Han, H.S. Choi, J.G. Kim. "Meta-modeling of nonlinear impulsive dynamics using composite neural network model with differential propagation", *Conference on Engineering Reliability, Korean Society of Mechanical Engineers* (**Excellent Paper Award**, Oral Presentation). *Extended works from IMAC 2023 presentation.* [link] — May 18 2023, Busan, South Korea

H. Lee, S. Han, H.S. Choi, J.G. Kim. "Composite neural network framework for modeling impulsive nonlinear dynamic responses", *41th International Modal Analysis Conference (IMAC)* (Oral Presentation). [link] — Feb. 16 2023, Austin, TX, USA

H. Lee, J.G. Kim. "Development of multibody dynamics trailer model using normal transportation test data and DNN based surrogate model generation", *Fall conference, Korean Society for Noise and Vibration Engineering* (Oral Presentation). — Dec. 4 2022, Jeju, South Korea

SKILLS

- **Programming Languages:** Python · MATLAB · C# · C++ · LaTeX
- **Software & Frameworks:** PyTorch · Git · Docker · ROS · RecurDyn · Gazebo · MuJoCo · OpenAI API
- **Scientific Computing:** Flexible multibody dynamics · Finite element methods · Numerical analysis
- **Natural Languages:** Korean (Native) · English (TOEFL iBT 111) · Japanese (Advanced proficiency)

SELECTED PROJECTS

- Segment-Anybody, at *Kyung Hee University Dept. of AI.* [GitHub]
  - Fine-tuned Meta's Segment-Anything Model (SAM) for detailed human body parts segmentation in human-robot interaction. Doubled its IoU scores on the public PASCAL-VOC 2010 dataset.
- Deep seq2seq from scratch, *Personal project.* [GitHub]
  - Scratch-implemented deep sequence-to-sequence models, from RNN to Transformer variants.
- ROS SLAM guide for Apple Silicon devices, *Personal project.* [GitHub]
  - Addressed ROS TurtleBot3 SLAM and Gazebo compatibility issues on the latest Mac devices.
- RecurDyn ProcessNet with Python, at *Modeling & Simulation Lab.* [GitHub]
  - Automated and parallelized flexible multibody dynamics (FMBD) simulations of RecurDyn.

TEACHING & MENTORSHIP

Research Mentor — Modeling & Simulation Lab., Kyung Hee University, Mar. 2025 — Present
— *Guiding a junior M.S. student (Minjae Jung) on time series analysis and modeling.*

Teaching Assistant, Multibody System Dynamics — Dept. of Mechanical Eng., Kyung Hee University, Mar. 2022 — Jun. 2023
— *Lectured on computational multi-body dynamics with theoretical derivations.*

LEADERSHIP & SERVICE

Lead Administrative Assistant — Dept. of Mechanical Engineering, Kyung Hee University, 2022 — 2024
College of Engineering 48th Student Council — Kyung Hee University, Feb 2019 — Jan. 2020
U.S.-R.O.K. Combined Marine Corps Interpreter — 1st Marine Div., ROKMC, Sep 2017 — Feb. 2019